FACULTY OF INFORMATION TECHNOLOGY CTU IN PRAGUE

Assignment of bachelor's thesis

Title:	Enhancing Fake News Classification with Advanced NLP Models
Student:	Nikita Bardatskii
Supervisor:	Ing. Mgr. Ladislava Smítková Janků, Ph.D.
Study program:	Informatics
Branch / specialization:	Artificial Intelligence 2021
Department:	Department of Applied Mathematics
Validity:	until the end of summer semester 2026/2027

Instructions

The aim of this thesis is to build upon previous research in fake news classification and extend it by incorporating advanced natural language processing models (e.g., transformer architectures such as BERT, GPT, or their variants) or other specific architectures used for such tasks. If needed, the existing datasets may be expanded with relevant content obtained from online sources. The student will conduct experiments with various text preprocessing configurations and advanced models, evaluate their accuracy and robustness, and compare them with the basic methods used in the original work. Emphasis will be placed on clearly describing the student's own contributions (i.e., dataset extension, implementation, and interpretation of new models).

1. Research current state-of-the-art techniques.
2. Implement selected methods using advanced and specialized models BERT, GPT, or other transformer networks. Implement at least 3 methods.
3. Make experiments. Present experimental results clearly and compare them with previous work.

Supporting documents will be provided by the thesis supervisor.

Electronically approved by Ing. Magda Friedjungová, Ph.D. on 18 March 2025 in Prague.